Multi-Instance Object Recognition in Cluttered Scenes

Assignment 1 — Computer Vision

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1 Object and Data

The object chosen for this assignment is a flat printed card (an "ALL DRUNK" branded card with dense black-and-white doodle graphics and typography). Seven physical copies of the card, in the form of stickers and the card itself, were placed across a cluttered study table at different positions, orientations, and scales, with several partially occluded by other objects.

- **Template image**: an isolated, flat photograph of the card.
- **Query image**: a single cluttered photograph of the desk containing multiple instances of the card.

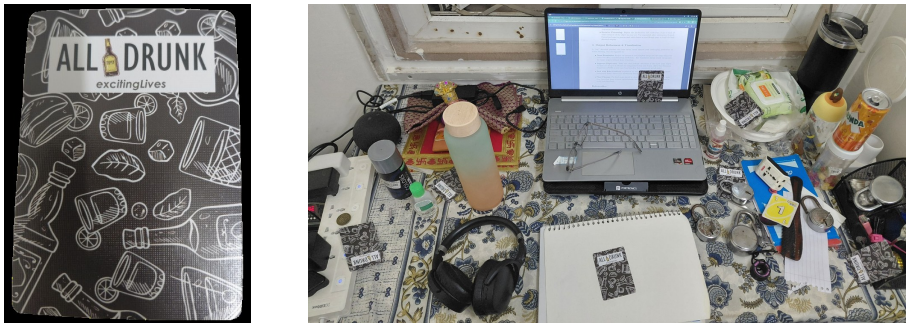


Figure 1: Left: isolated template image. Right: cluttered query image containing multiple instances of the object.

2 Recognition Pipeline: Mathematical Implementation

The pipeline follows five stages: SIFT feature extraction (OpenCV), feature correspondence, 4D Generalized Hough Transform (GHT) clustering, RANSAC-based affine estimation, and duplicate suppression. Only the SIFT keypoint/descriptor extraction step uses a high-level OpenCV routine (`cv2.SIFT_create`); every subsequent stage is implemented directly.

2.1 SIFT Feature Extraction

Both images are converted to grayscale, and the scene image is resized so that its width does not exceed `max_width = 2400` px (to bound memory and runtime). OpenCV's SIFT detector then returns, for every keypoint k, a location (x_k, y_k), a scale s_k, an orientation $\theta_k \in [0°, 360°)$, and a 128-dimensional descriptor vector $\mathbf{d}_k$.

2.2 Feature Correspondence (Lowe's Ratio Test)

For every scene descriptor $\mathbf{d}_s$, its Euclidean distance to every template descriptor $\mathbf{d}_t$ is computed using the vectorized identity

$$\|\mathbf{d}_s - \mathbf{d}_t\|^2 = \|\mathbf{d}_s\|^2 + \|\mathbf{d}_t\|^2 - 2\,\mathbf{d}_s \cdot \mathbf{d}_t, \tag{1}$$

evaluated for a whole chunk of scene descriptors against all template descriptors at once via matrix multiplication. Let d_1 and d_2 be the smallest and second-smallest such distances for a given scene descriptor. The match is accepted only if

$$\frac{d_1}{d_2} < 0.75 \quad \text{(RATIO_THRESHOLD)}. \tag{2}$$

Matching is performed *scene* $\rightarrow$ *template* (each scene keypoint searches for its nearest template keypoint) rather than the reverse, so that several scene instances can each independently match the same template keypoint.

2.3 4D Generalized Hough Transform

Every accepted match (k_s, k_t) casts one vote for the object's pose in the scene. Let $(c_x, c_y) = (w_t/2,\, h_t/2)$ be the centre of the template image, and let $(dx, dy) = (c_x - x_{k_t},\, c_y - y_{k_t})$ be the vector from the template keypoint to that centre. The predicted scale, rotation, and centre location are

$$\text{scale} = \frac{s_{k_s}}{s_{k_t}}, \tag{3}$$

$$\theta = (\theta_{k_s} - \theta_{k_t}) \bmod 360^\circ, \tag{4}$$

$$\text{pred}_{c_x} = x_{k_s} + \text{scale} \cdot (dx \cos\theta - dy \sin\theta)\,, \tag{5}$$

$$\text{pred}_{c_y} = y_{k_s} + \text{scale} \cdot (dx \sin\theta + dy \cos\theta)\,. \tag{6}$$

Each vote $(\text{pred}_{c_x}, \text{pred}_{c_y}, \text{scale}, \theta)$ is discretized into a 4-D bin using bin widths of 50 px (position), 0.1 (scale), and 30° (rotation), and accumulated in a hash map (rather than a dense array, since the occupied region of the 4-D space is extremely sparse). Bins with fewer than `HOUGH_MIN_VOTES = 3` votes are discarded; the remaining bins are the candidate object poses.

2.4 Robust Affine Estimation (RANSAC + Least Squares)

For each candidate bin, a 2-D affine model

$$\begin{bmatrix} x_s \\ y_s \end{bmatrix} = A \begin{bmatrix} x_t \\ y_t \end{bmatrix} + \mathbf{t}, \qquad A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \tag{7}$$

is fit robustly:

1. Three correspondences are sampled at random. If the three template points are (nearly) collinear, i.e. their triangle has twice-area below a threshold, the sample is discarded.

2. The six parameters (a, b, c, d, t_x, t_y) are solved from the linear system $M\mathbf{x} = \mathbf{v}$, built by stacking, for every correspondence, the two rows

$$\begin{bmatrix} x_t & y_t & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & x_t & y_t & 1 \end{bmatrix}, \tag{8}$$

   and solved via least squares (`numpy.linalg.lstsq`); for a minimal 3-point sample this is an exact solve, and for the final refit over all inliers it is a genuine least-squares fit.

3. The fitted A is rejected as degenerate if any of the following hold: (i) $|\det A| < 10^{-8}$ (collapse to a line/point); (ii) the ratio of the norms of the two columns of A (the images of the template's axes) exceeds 3.0 (anisotropic stretch); or (iii) the angle between those two columns deviates from 90° by more than 35° (excessive shear). This test does not assume any particular absolute scale, so it accepts instances that are both much smaller and much larger than the template.

4. Every correspondence in the cluster is reprojected through the candidate model, and counted as an inlier if its reprojection error is below 5 px. The sample with the largest inlier count, over 1000 random trials, is kept, and the affine model is then re-estimated using *all* of its inliers.

2.5 Acceptance Criteria and Iterative Multi-Instance Extraction

A verified affine model is accepted as a detected instance only if all of the following hold:

- inlier count ≥ 3;
- inlier ratio $= \dfrac{\text{inliers}}{\text{cluster size}} \geq 0.4$;
- mean reprojection error over the inliers ≤ 3 px.

Hough bins are processed in decreasing order of vote count. Once a bin's model is accepted, its inlier matches are permanently removed from the pool of available correspondences before the next bin is processed (sequential inlier subtraction), so that the same physical instance cannot be re-detected by a different, overlapping bin, and iteration continues until every remaining bin has been processed.

2.6 Duplicate Suppression via Polygon IoU

Each accepted model projects the template's four corners into the scene, giving a (possibly rotated) quadrilateral. Because instances may be rotated, Intersection-over-Union is computed on general polygons rather than axis-aligned boxes:

- the area of a simple polygon is computed with the shoelace formula, $\text{Area} = \frac{1}{2}\left|\sum_i x_i y_{i+1} - y_i x_{i+1}\right|$;
- the intersection polygon of two quadrilaterals is computed with the Sutherland–Hodgman clipping algorithm, implemented directly (including its own line-intersection routine);
- $\text{IoU} = \text{Area}_\cap / \left(\text{Area}_A + \text{Area}_B - \text{Area}_\cap\right)$.

Detections are kept greedily; a candidate is discarded as a duplicate of an already-kept detection if their IoU exceeds 0.3.

3 Results: Naive Matching vs. Final Detections

Table 1 summarizes the pipeline's intermediate and final counts on the query image in Figure 1, together with the range of verification statistics across the bins that were accepted.

Quantity	Value
Template keypoints	2783
Scene keypoints	8000
Raw matches after ratio test	397
Hough bins with ≥ 3 votes	19
Verified instances before NMS	19
Final instances after NMS	**7**
Inlier ratio (accepted bins)	0.94 – 1.00
Mean reprojection error (accepted bins)	0.00 – 1.71 px

Table 1: Summary statistics for the pipeline run shown in Figures 2–3.

Note that 19 candidate bins were independently verified, but only 7 survive NMS: several Hough bins re-discover the same physical card from slightly different vote clusters, and the polygon-IoU NMS step correctly collapses these into one detection per real object instance.

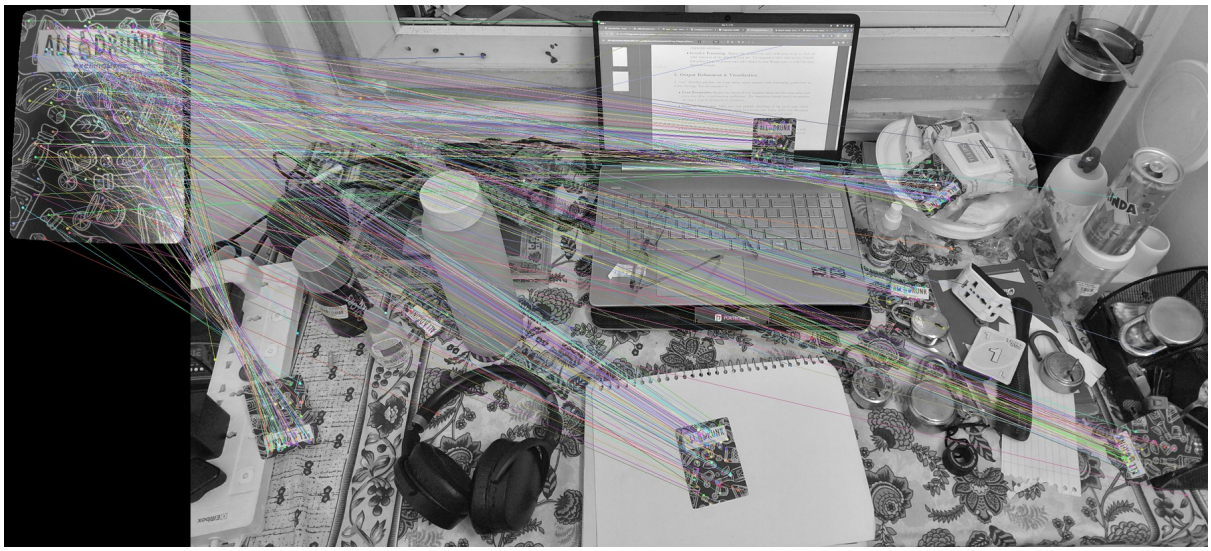


Figure 2: Naive matching: every correspondence that passed Lowe's ratio test, drawn without any geometric verification. Correct matches converge on the seven real card instances, alongside stray matches to unrelated background clutter (keyboard, cables, notebook edges).

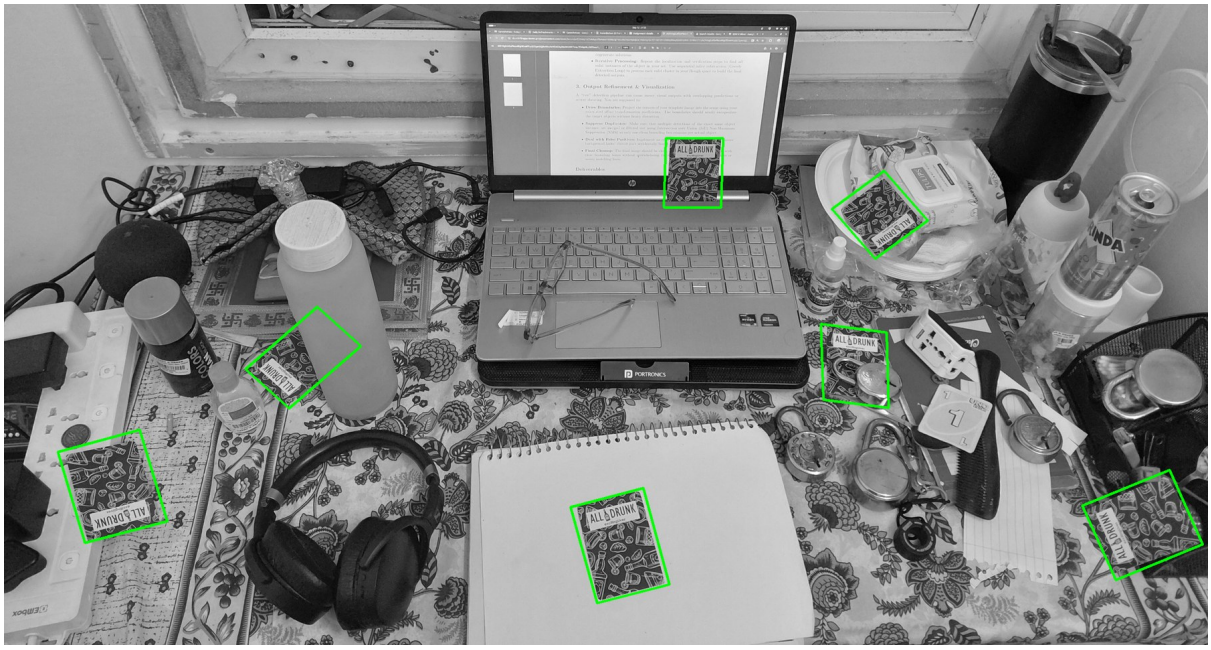


Figure 3: Final detections after 4D GHT clustering, RANSAC affine verification, and polygon-IoU NMS: seven clean, correctly oriented bounding quadrilaterals, one per real object instance, with no stray keypoints or false positives.